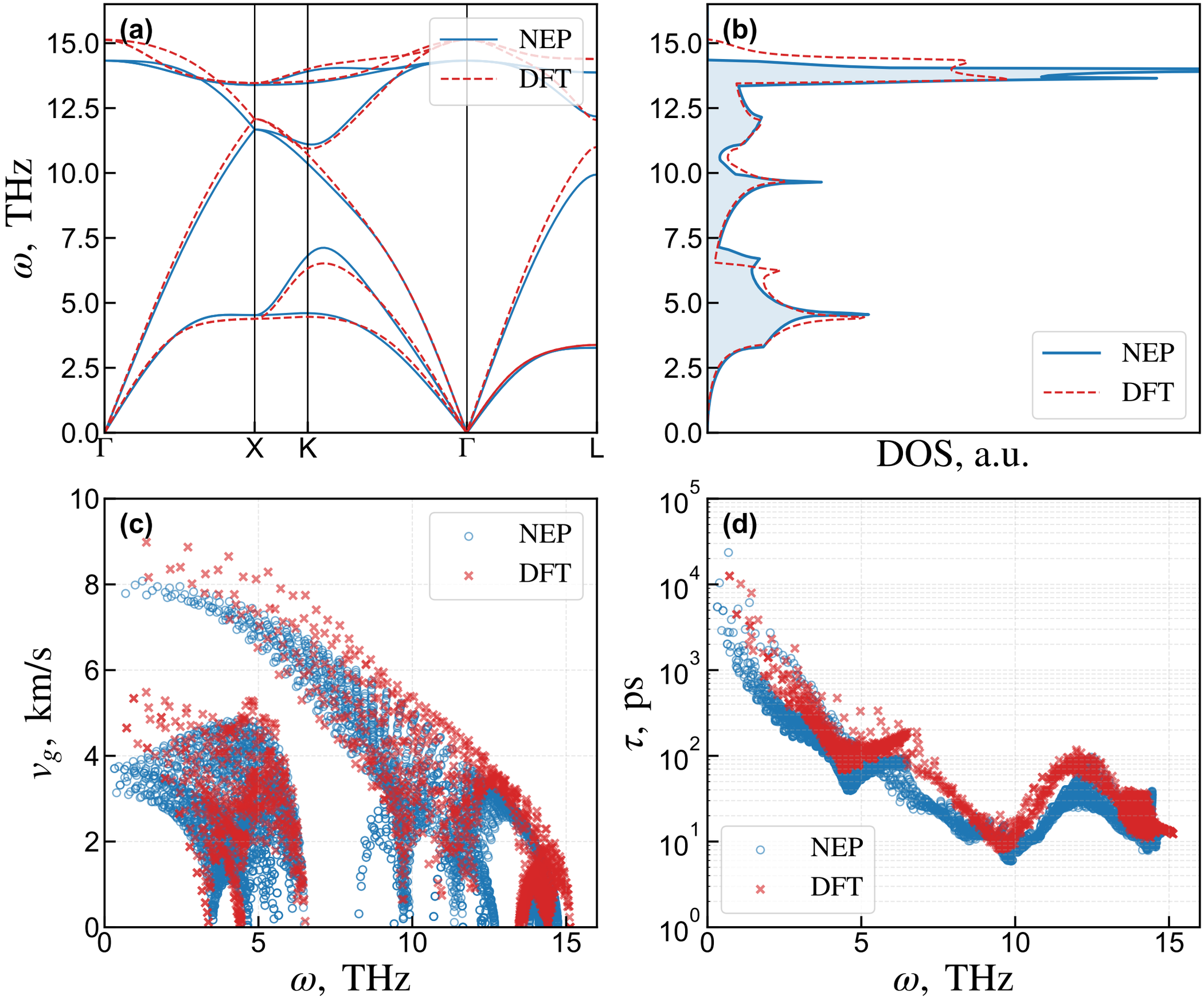
(a)
NEP
DFT
ω, THz
Γ
X
K
Γ
L
(b)
NEP
DFT
DOS, a.u.
(c)
NEP
DFT
v_g, km/s
ω, THz
(d)
NEP
DFT
τ, ps
ω, THz